

Git: A Popular DevOps Tool

Welcome to the world of Git! If you're new to software development, understanding Git is like learning a new language — essential for effective collaboration and project management.



Git is like a time machine for your projects, allowing you to revisit previous versions, experiment with new ideas, and work seamlessly with others. In this guide, we'll give a clear explanation of Git's purpose and core concepts and understand its benefits for both individual and team projects. We will discuss essential terminology and workflows to kickstart your Git journey, and learn how to install, configure, and create your first repository.

What is Git?

Git is essentially a tool that helps you track changes in your code over time. Imagine you're writing a novel. You start with a blank page and gradually add chapters, characters, and plot twists. Without a backup or a detailed record of your writing process, it would be difficult to go back and change something without affecting the entire story.

Git is like a digital notebook for your code. It records every change you make, allowing you to:

- **Revert to previous versions:** If you introduce a bug, you can go back to a working version and figure out what went wrong.
- **Experiment freely:** You can try out new features without affecting the main project.
- **Collaborate with others:** Multiple people can work on the same codebase without overwriting each other's changes.

Many software developers are often confused between Git and GitHub. Table 1 compares the two to clear this confusion.

As we know, Git is a powerful version control system (VCS) that is essential for modern software development. Table 2 lists key reasons why Git is important for both individual and team projects.

What are the core concepts of Git?

Git is built on a few fundamental concepts that are essential to understanding how it works.

Repository (Repo): A Git repository is a storage space for your project. It holds all your project's files and the history of their changes. It can be local (on your own machine) or remote (on a server like GitHub or GitLab).

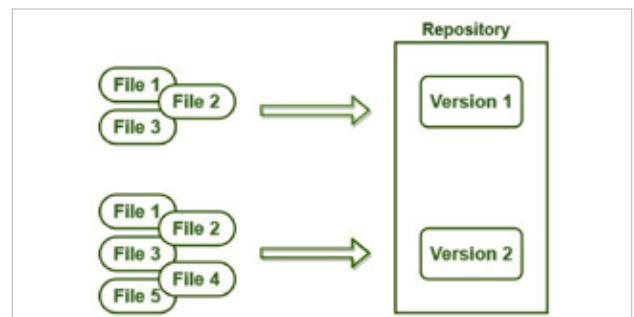


Figure 1: Repository (Repo)